

Standardizing sablefish (*Anoplopoma fimbria*) long-line survey abundance indices by modeling the log-ratio of paired comparative fishing cpues

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From 1979 to 1994, the US National Marine Fisheries Service (NMFS) and the Fisheries Agency of Japan (FAJ, 1979–1988), and later the North Pacific Cooperative Fisheries Company (NPCFC, 1989–1994), have performed joint long-line surveys on chartered Japanese commercial fishing vessels in the Gulf of Alaska. These surveys were designed primarily to monitor the relative abundance of sablefish (*Anoplopoma fimbria*). To prepare for the future termination of these joint surveys, between 1988 and 1994 NMFS performed similar long-line surveys aboard US vessels in the Gulf of Alaska. These paired (i.e. duplicated) surveys from 1988 to 1994, provide a basis for comparing future NMFS survey results with the past time series of joint US and Japanese surveys (i.e. the joint 1979–1994 surveys). This paper describes how modeling the log-ratio of paired cpue results (i.e. the $\log[(\text{NMFS cpue})/(\text{joint cpue})]$), from a standardized survey design, was used to provide a basis for converting the joint survey cpue values to values directly comparable to those from the NMFS surveys. This was necessary because sablefish long-line surveys will be carried out solely by NMFS in the future.

Key words: linear models, functional regression models.

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Introduction

Joint long-line surveys were performed in the Gulf of Alaska by the Alaska Fisheries Science Center (AFSC) of the US National Marine Fisheries Service (NMFS), and the Japanese Far Seas Fisheries Institute of the Fishery Agency of Japan (FAJ) from 1979 to 1988 (Sigler and Zenger, 1989). The primary purpose of these surveys, performed by Japanese and US scientists aboard Japanese commercial long-line vessels chartered by the FAJ, was to track the annual abundance of sablefish (*Anoplopoma fimbria*) by monitoring catch rates at standard sampling locations. Following the 1988 survey, the Japanese government withdrew its participation in the joint surveys due to termination of foreign long-line fishing in US waters.

In 1989, the North Pacific Cooperative Fisheries Company (NPCFC), a private Japanese company, agreed to continue the annual joint surveys in return for permission to retain most of the catch to defray charter expenses. When FAJ withdrew from these surveys, it

was unclear how long the joint surveys would continue. To prepare for the eventual termination of joint surveys, NMFS (i.e. the AFSC) conducted their own annual long-line surveys in the Gulf of Alaska from 1988 through 1994. These NMFS surveys fished the same standard locations (Fig. 1) as were used in the joint surveys. The comparative fishing data provided from these paired NMFS and joint surveys allow the standardization (or conversion) of the historic joint survey data series, to a series directly comparable to the NMFS surveys. This is important because future surveys will be conducted solely by NMFS.

Gulland (1956) and Beverton and Holt (1957) recognized that if catch per unit effort (cpue) data were to reflect the relative abundance of fish stocks, then standardization would be required to take into account variability in the ability of fishing vessels to catch fish. The method of analysing $\log(\text{cpue})$ with linear models was originated by Gulland (1956), expanded on by Robson (1966), and is now the standard method for analysing cpue data. More recent papers (Gavaris, 1980;

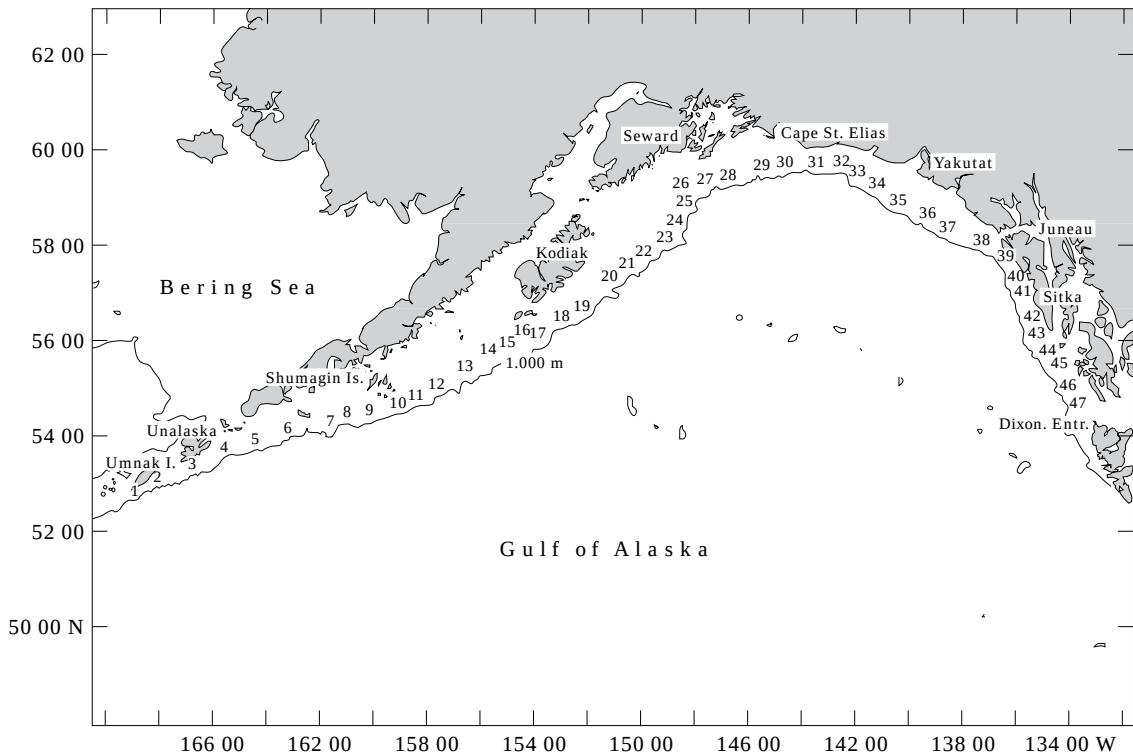


Figure 1. Station locations for the joint and NMFS long-line surveys.

Kimura, 1981, 1988; Stocker and Fournier, 1984) have emphasized the generality of the method and provided additional mathematical insights.

Our methods differ from these more traditional methods in several important respects. First, it is assumed that only two fishing gear types were involved, and that annual surveys were conducted using a common, carefully detailed survey design. Second, we assume that in some year(s) both fishing gears replicated the survey design, providing a basis for standardizing results from the two gears. For the purposes of discussion, we consider the replicated survey cpues resulting from a particular year, and survey design stratum to be *paired* results. Finally, the goal of our modeling and analysis is to *convert* one gear type's survey cpue, for each year and survey design stratum, to a value that is, to the greatest extent possible, directly comparable with that of the other gear. Major advantages of this approach are that converted cpue estimates are available for each year-stratum so they may be summarized as desired, and that future survey results can be compared with past results without further standardization or analysis of data.

We chose to model the log-ratio of the observed cpue from each gear type, because the model and its application are conceptually straightforward. Suppose that a design matrix, X , containing only survey design

variables (e.g. site, depth, and size), explains the observed log-ratios, Y . An analysis of variance based on X will highlight those factors for which the relative fishing power of the two gears varies significantly. Next, we fit a linear model which includes these significant factors. Finally, given the cpue from gear 2 and its corresponding design matrix, we use the fitted model and the design matrix to estimate the log-ratio of cpues, then estimate the cpue from gear 1 using the estimated log-ratio and the observed values of the cpue from gear 2. In this way we can convert observed gear 2 cpues to predicted gear 1 cpues for years in which gear 1 cpues were not observed. In our analysis, gear 2 cpue is the joint sablefish survey long-line cpue extending into the past, and gear 1 cpue is the NMFS sablefish long-line survey cpue which will extend into the future. The goal will be to convert the observed joint survey cpues to estimated NMFS survey cpues so that they can be compared with future NMFS survey cpue observations.

It is generally understood that the main barrier to accomplishing this goal is the existence of interaction between year effects and either gear definition variables or stratum definition variables (Large, 1992). Interaction between year effects and gear effects can be detected as either a year:gear interaction effect (we use ":" to denote interactions) in an analysis of variance (ANOVA) of $\log(\text{cpue})$, or as a year effect in an ANOVA of log-ratios.

A significant year:gear interaction indicates that the cpue for the different fishing gears show different trend in abundance across years. Further discussion of this common situation can be found in our conclusions.

The existence of interaction between year and stratum definition effects indicates that the different strata may show different trends in abundance, and may benefit from being treated separately. However, such interactions do not preclude averaging over the different strata to get an average trend in abundance.

Methods

Survey methods and materials

Vessels and sampling gear

Various Japanese commercial fishing vessels, 50–55 m in length, were chartered for joint surveys. The NMFS surveys used the 35 m *F/V Prowler* in 1988, the 47 m *F/V Ocean Prowler* from 1989 to 1993, and the 47 m *F/V Alaskan Leader* in 1994.

The Japanese long-line gear used in the joint surveys and the NMFS long-line gear are similar in many respects (Fig. 2; see Zenger, 1995). At each station, both surveys deployed 9.5 mm diameter ground-lines composed of 160 skates (sections of long-line gear tied together). Each skate is 100 m long and contained 45 hooks. Hooks were spaced 2 m apart. A 3 kg weight was attached where the skates were joined. Sixteen kilometres of groundline with 7200 hooks were fished at each station. Both surveys used squid (*Illex* spp.) as bait.

Differences between Japanese sampling gear and NMFS sampling gear were mostly due to hook and gangion construction. The joint survey (i.e. Japanese gear) used J-shaped (tara) hooks and 1.2 m gangions made of #48 twisted nylon tied to the ground-line. The NMFS hooks and gangions were generally more heavily constructed than those of the joint survey. The NMFS survey used circle hooks and 38 cm gangions that were secured to beackets tied into the groundline. Gangions were constructed of #60 braided nylon. During the 1988 NMFS survey and for part of the 1989 NMFS survey, beacket material was #60 braided nylon, but during the 1989 survey all beackets were changed to #72 braided nylon. No changes have been made to the NMFS sampling gear specifications since 1989. Daily gear maintenance assured that sampling effort remained as constant as possible.

In this paper “gear” will be synonymous with survey type. Survey cpue1 is NMFS (1988–1994) long-line cpue using gear 1, while survey cpue2 is joint (1979–1994) long-line cpue using gear 2. Our analysis will not distinguish between the FAJ and NPCFC joint surveys.

Survey area and operation

The survey area extended along the upper continental slope between Umnak Island (52°50'N 170°W) eastward

to Dixon Entrance (Fig. 1). Sampling depth ranged from approximately 200 m to 1000 m. Standard sampling locations were specified in the sampling design.

Both the joint and NMFS surveys recorded catch frequency by species for each skate of gear, depth at each fifth skate or at change of depth stratum, and extensive size composition data measured in fork length to the nearest cm. Depth strata were: 201–300 m, 301–400 m, 401–600 m, 601–800 m, and 801–1000 m. Ineffective hooks (i.e. those that were broken, bent, tangled or missing) were tallied by skate for the NMFS survey.

Cpue calculations

The average number of sablefish caught per skate (i.e. cpue) were calculated for each year, station-depth stratum. These data were used in the functional linear regression analysis described later.

The NMFS catch rates were standardized to catch per 45 hooks for skates with up to 5 ineffective hooks. Skates with 6 or more ineffective hooks were not used. No effort correction was made in calculating joint survey cpue data, but catches from skates that were damaged were not used.

For the ANOVA and models associated with the standardization of data, cpue (i.e. catch in numbers per skate) were calculated by year, location, depth, and fish size category. Strata were combined to avoid empty cells in the data analysis. Locations were redefined as: site 1=stations 1 and 2, site 2=stations 3 and 4, etc. New depth strata were defined as: new depth 1=201 m–400 m, new depth 2=401 m–600 m, and new depth 3=601 m–1000 m. Sablefish size categories were: small ≤ 56 cm, medium = 57–66 cm, and large ≥ 67 cm.

Statistical methods

Linear functional regression

Functional regression allows us to explore the relationship between mathematical variables that are masked by measurement error. If we assume that our observed cpues are linearly related to “true” cpue (i.e. abundance), then the relationship between the joint and NMFS surveys can be shown to have a linear functional relationship (Kimura, 1992a). When the ratio of the variances of the measurement errors is known to be $\lambda = 1.0$ (i.e. the measurement error is the same magnitude for the two surveys), the maximum likelihood estimates are well-known (Kendall and Stuart, 1973; Kimura, 1992b) and are often referred to as the “ordinary” major axis (MA), and the method of estimation referred to as principal axis regression. However, functional regression does not appear to provide a suitable avenue for standardizing surveys. This is because it is not easy to incorporate design variables into the relationship, and more importantly, the estimated functional relationship is between posited “true” values and not the observed

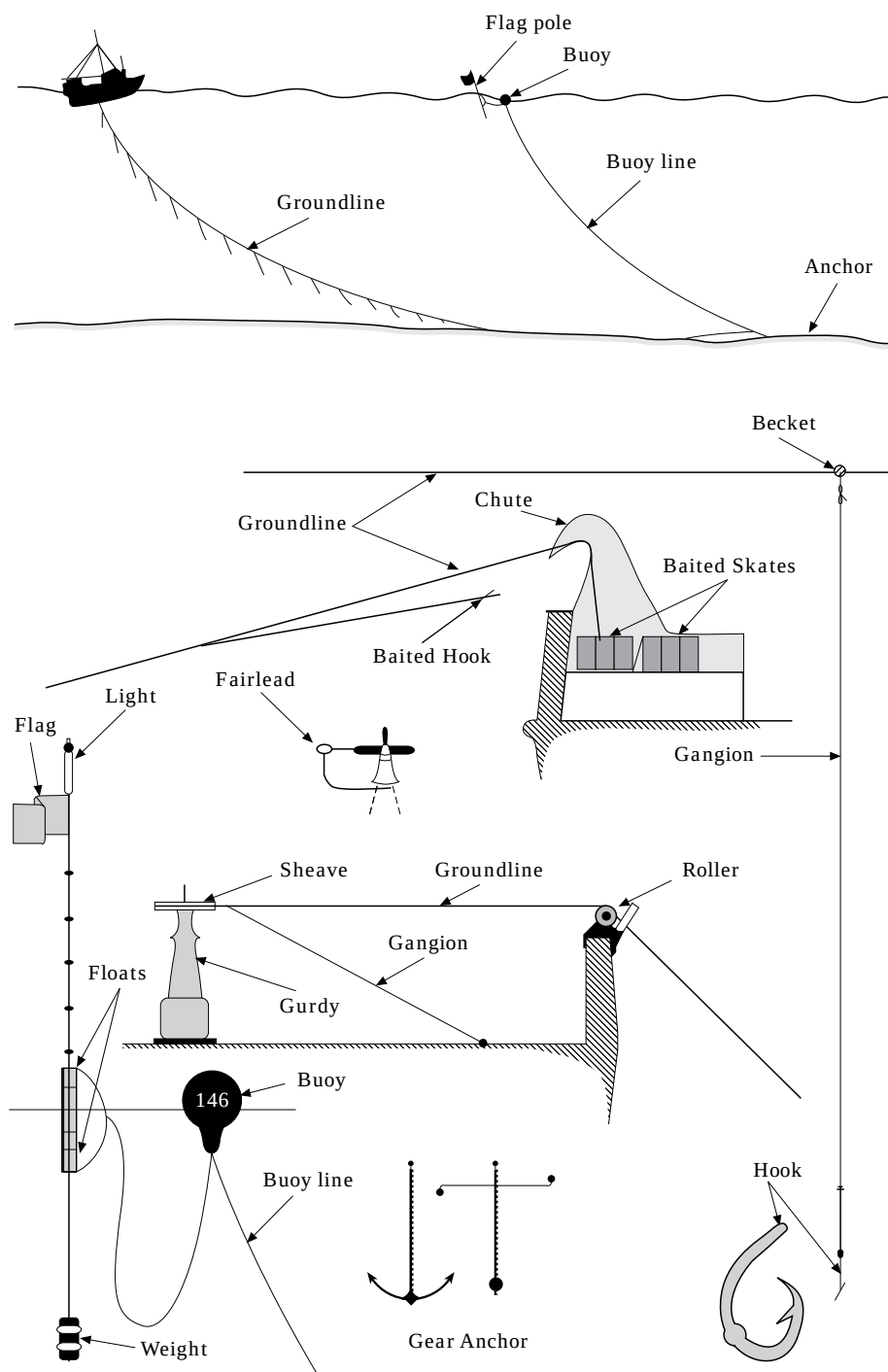


Figure 2. Drawing of long-line fishing gear and hauling equipment (from Charles R. Hitz, NMFS).

values that we have to work with. We believe the ordinary regression models described below are more appropriate for actual survey standardization (i.e. the prediction problem).

Abundance estimates from the log-ratio model

The statistical method used in this paper is the standard one of analysing a log-transformed response variable using linear models.

Let

i =the year index,

j =the gear index,

k =the site index,

l =the depth index,

m =the size index,

c_{ijklm} =the catch in numbers,

e_{ijklm} =the effort expended in obtaining the catch, and

$u_{ijklm}=c_{ijklm}/e_{ijklm}$ be the cpue.

The dependent variables, all column vectors of dimension n , consist of paired observations fished under similar conditions (i.e. the same year-stratum),

$$Y_{g1} = \{\log(u_{i1klm})\},$$

$$Y_{g2} = \{\log(u_{i2klm})\},$$

$$Y = Y_{g1} - Y_{g2},$$

$$= \{\log(u_{i1klm}) - \log(u_{i2klm})\},$$

$$= \{\log(u_{i1klm}/u_{i2klm})\},$$

=the log-ratio of cpues.

To determine the factors affecting the relative fishing power between the NMFS and Japanese long-line gears, we restrict our analysis to years when the surveys were replicated, and perform an analysis of variance on log-ratios for all main effects and two-way interactions.

$$\log(u_{i1klm}/u_{i2klm}) = \mu + \alpha_i + \sigma_k + \delta_l + \rho_m + (\alpha\sigma)_{ik} + (\alpha\delta)_{il} + (\alpha\rho)_{im} + (\sigma\delta)_{kl} + (\sigma\rho)_{km} + (\delta\rho)_{lm} + \varepsilon_{ijklm}$$

The factors are μ =mean, α_i =year, σ_k =site, δ_l =depth, and ρ_m =size. Because the response variable is a log-ratio, our factors actually represent two-way and three-way interactions with the gear factor (e.g. a significant year main effect actually means the year:gear interaction is significant). After determining which factors most affect relative fishing power, we can then determine which factors and interactions should be included in the standardization models that will be used to convert joint cpues to values comparable to observed NMFS cpues. At this stage, we must discard year and year interaction effects because it is not possible to use these factors to standardize future or past data.

The linear model for converting cpue2 to cpue1 is arrived at by fitting the log-ratios to the factors selected by the analysis of variance using ordinary least squares. Suppose in addition to a comparative fishing dataset (Y, X) , there exists a historical (or “future”) dataset $(\{u_{i2klm}\}, M_i)$ of sample data using gear 2. Here, M_i is the design matrix corresponding to observations of $\{u_{i2klm}\}$. We change notation from X to M_i because X is the design matrix of the log-ratio data selected to estimate model parameters and contain no year subscripts, but M_i is different for every year of available data and is used only for the prediction of log-ratios. The regression

coefficients estimated from (Y, X) can then be used to estimate log-ratios $\hat{Y}_i$ from M_i . The unobserved $\{u_{i1klm}\}$ can then be estimated from the elements of $\exp(\hat{Y}_i)$ by multiplying by the observed u_{i2klm} . We then have a full time series of estimated $\{\hat{u}_{i1klm}\}$ that can be suitably summarized as standardized estimates of relative abundance. Note that the analysis described here can be used to estimate standardized indices in either of the fishing gears based on either past or “future” data.

More formally, let the model for estimating log-ratios be $Y = X\beta$, and let $\Sigma(\cdot)$ denote the covariance matrix of any vector argument. Then $\hat{\beta} = (X'X)^{-1}X'Y$ and $\hat{\Sigma}(\hat{\beta}) = (X'X)^{-1}s^2$ where s^2 is the estimate of residual variance.

Suppose we wish to estimate the cpue1 = $\{u_{i1klm}\}$ average index, in a year for which only a design matrix M_i and the corresponding vector of cpue2 = $\{u_{i2klm}\}$ indices are available. First estimate the predicted log-ratio vector $\hat{Y}_i = M_i\hat{\beta}$, and its covariance matrix $\hat{\Sigma}(\hat{Y}) = M_i(X'X)^{-1}M_i's^2$. Define $\hat{R}_i = \exp(\hat{Y}_i)$ as the estimated ratios of cpues. Then the estimated column vector of predicted cpue1 indices is $\hat{cpue1} = \{\hat{R}_i \cdot cpue2_i\}$. The covariance matrix is

$$\hat{\Sigma}(\hat{cpue1}) \approx \{\hat{cpue1}_i \hat{\Sigma}(\hat{Y})_{ij} \hat{cpue1}_j + \hat{R}_i \hat{\Sigma}(\hat{cpue2}_{ij}) \hat{R}_j\}$$

which is derived by the delta method and subscripts i, j refer to entries in the corresponding vector or matrix. The first component is covariance caused by standardization, while the second component is the covariance due to sampling cpue2 (Appendix 1). Let $c' = (1/n, \dots, 1/n)$ be a vector of length n where cpue1 is a vector of length n . Then we define the average abundance index as $\bar{a}_i = c' \hat{cpue1}$ with $\text{var}(\bar{a}_i) = c' \hat{\Sigma}(\hat{cpue1}) c$. We used these formulas to estimate gear 1 relative abundance indices for all years, including those years for which gear 1 sampling actually occurred. In this way, we hope to reduce bias in the inter-annual comparisons of abundance.

Abundance estimated from the year-effects model

Another method of arriving at standardized estimates of relative abundance is to fit $\log(\text{cpue})$ to the year effect, gear effect, and other appropriate terms again using linear models. The rationale being that the year effect, which can be interpreted as an index of relative abundance (Kimura, 1981), will be standardized with respect to the variables entered into the model. Here, the observed variables would be $\{Y_{g1}\}$ and $\{Y_{g2}\}$, not necessarily paired; and the design matrix would contain the year effect, gear effect and other appropriate terms. We shall call this the year-effects model, and the estimates of relative abundance estimated from this model the year-effects estimates.

The main problem that arises is that of model selection. For the log-ratio analysis, the problem of model

selection was based simply on which variables were most significant in an ANOVA of log-ratios. Similarly, we performed an analysis of variance of log(cpue) using the year, gear, site, depth and size factors, and all two factor interactions. More terms were not included because of limitations of computer memory, missing values, and the desire for parsimony.

Theoretically, the log(cpue) linear model should include the year effect, gear effect, gear interaction terms for factors that affect relative fishing power, and factors and interactions that appear statistically significant. We found such complex models became non-estimable probably due to missing observations. The approach we will use is just to include the year effect, gear effect, and site, depth and size effects nested in gear:

$$\log(u_{ijklm}) = \mu + \alpha_i + \beta_j + (\beta\sigma)_{k(j)} + (\beta\delta)_{l(j)} + (\beta\rho)_{m(j)} + \varepsilon_{ijklm}.$$

The factors are μ =mean, α_i =year, β_j =gear, σ_k =site, δ_l =depth, and ρ_m =size. We can then estimate relative abundance using the regression coefficient estimates for the year effect. The estimated indices of relative abundance based on the year effect are the $\exp(\hat{\alpha}_i)$ (Kimura, 1981). If annual relative abundance indices are to be made for specific categories, the regression coefficient indices must be included for these categories. For example, $\exp(\hat{\mu} + \hat{\alpha}_i + \hat{\beta}_1 + (\hat{\beta}\rho)_{2(1)})$ will provide relative abundance indices for year i , gear 1, and size category 2.

It is interesting to consider the difference between the log-ratio method and year-effects method of obtaining relative abundance indices standardized to the NMFS long-line survey. The log-ratio method explicitly converts the cpues from the joint long-line survey to numbers comparable to NMFS long-line results. The year-effects method essentially provides adjustments for gear and size within gear categories.

Bias in logarithmic models

It is well-known that transforming parameter estimates from a logarithmic scale to the original scale can lead to biased estimates on the original scale (Beauchamp and Olson, 1973). However, the nature and direction of these biases can sometimes seem confusing. Under the log-normal assumptions, the β parameters estimated on the logarithmic scale are always unbiased. Say that we are estimating a year effect α_i . Then on the original observation scale, $\exp(\alpha_i)$ should be the abundance index. However, because $\hat{\alpha}_i$ is normally distributed, the quantity $\exp(\hat{\alpha}_i)$ overestimates $\exp(\alpha_i)$, $E[\exp(\hat{\alpha}_i)] = \exp(\alpha_i + \sigma^2_{\alpha_i}/2)$. Thus one possibility is to adjust the estimate downward by multiplying $\exp(\hat{\alpha}_i)$ by $\exp(-\hat{\alpha}_i^2/2)$ (Kimura, 1981). The antilog of any linear function of β will also be similarly biased, and can be adjusted in the same way; that is, the antilog of the linear function $\exp(c'\beta)$ can be estimated as

$\exp(c'\hat{\beta} - c'\hat{\Sigma}(\hat{\beta})c/2)$. This latter formula seems appropriate for estimating bias in the antilog of the predicted log-ratios.

Another adjustment problem arises when modeling observed variables as a function of model parameters. On the logarithmic scale, observations can be modeled as $\hat{Y}_i = f(\beta_i) + \varepsilon_i$ where ε_i is distributed as $N(0, \sigma^2)$, and σ^2 is the residual variance on the logarithmic scale. This suggests that expected values of observations on the original scale would be $E[\exp(\hat{Y}_i)] = \exp(f(\beta) + \sigma^2/2)$. If $f(\beta) = c'\beta$ is linear, and β is estimated, perhaps the approximation $E[\exp(\hat{Y}_i)] = \exp(c'\hat{\beta} - c'\hat{\Sigma}(\hat{\beta})c/2 + \hat{\alpha}^2/2)$ would be more accurate. Note that the term involving σ^2 appears to be in the opposite direction as the adjustment for the antilog of parameter estimates described above. More theoretical discussions concerning bias correction are provided by Finney (1941), Bradu and Mundlak (1970), Beauchamp and Olson (1973), and Gavaris (1980).

Results

Examination of scatter plots and linear functional regression lines (MA regressions) show that the annual relationship between joint and NMFS surveys varied significantly during the 1988 through 1994 time periods (Fig. 3; $p < 0.001$, $\hat{F} = 21.4$, d.f. = 12, 1412). This difference may be due in part to a learning curve for the AFSC staff performing the NMFS surveys. Therefore, we decided on using only comparative data for the years 1990–1994 which was far more uniform ($p = 0.048$, $\hat{F} = 1.96$, d.f. = 8, 1012) to calculate standardized cpue indices. The joint surveys for 1983–1994¹ were standardized to the NMFS surveys using the 1990–1994 comparative data.

Analysis of variance was performed on the log-ratio and log(cpue) data for a model including all main effects and two factor interactions for the 1990–1994 data (Tables 1 and 2). The significance of the year effect in the log-ratio ANOVA and the year:gear interaction effect in the ANOVA of log(cpue) suggests the presence of inter-annual variability in relative gear performance. Because year effects, and other significant year interaction effects (i.e. year:site, year:depth and year:size) cannot be used to standardize survey results between years, they show weaknesses in our analysis. Note also that in the log-ratio analysis (Table 1) main effects and interaction effects actually represent higher order interactions with gear.

Eliminating the year factor and re-calculating the ANOVA (Table 1) highlighted the factors that should be included in our model used to predict log-ratios: site,

¹The complete 1979–1982 joint survey data were not available to us at this time.

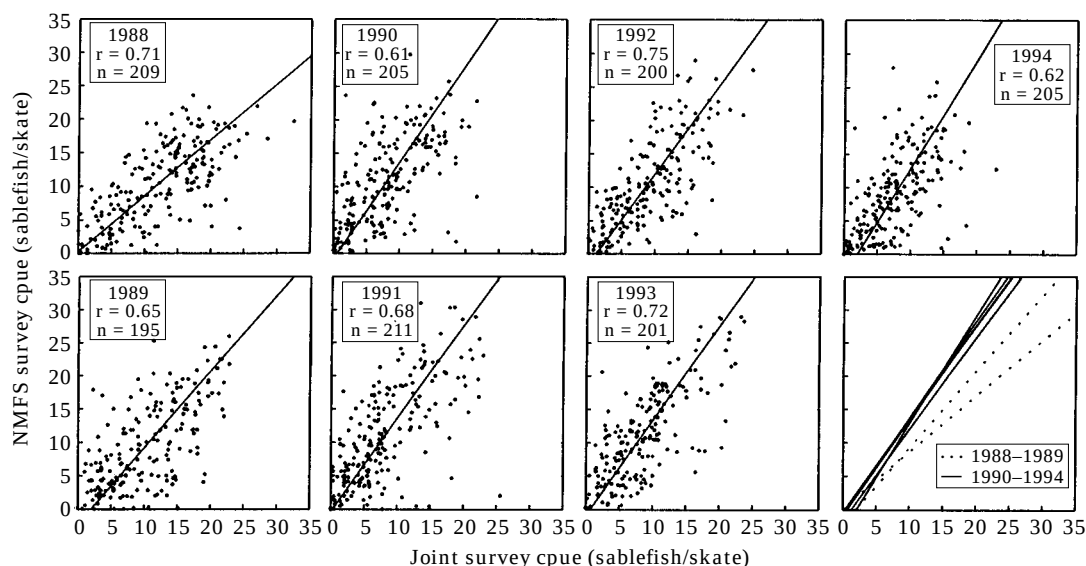


Figure 3. Functional linear regressions (i.e. major axis), by year, between joint and NMFS long-line survey data. Plot points are averages within station-depth strata (see text).

Table 1. Analysis of variance of log-ratios for sablefish longline cpue from the joint and NMFS long-line surveys (1990–1994). ANOVA includes all main effects and two-way interactions. Top table includes the year factor; bottom table excludes the year factor.

ANOVA including “year” factor					
Factor	Degrees of freedom	Sum of squares	Mean square	F-value	p-value
Year	4	7.108	1.777	6.79	0.0000
Site	23	44.264	1.925	7.36	0.0000
Depth	2	0.220	0.110	0.42	0.6571
Size	2	2.352	1.176	4.49	0.0114
Year:site	92	81.940	0.891	3.40	0.0000
Year:depth	8	3.043	0.380	1.45	0.1705
Year:size	8	11.157	1.395	5.33	0.0000
Site:depth	46	58.029	1.261	4.82	0.0000
Site:size	46	27.386	0.595	2.28	0.0000
Depth:size	4	1.810	0.452	1.73	0.1415
Residuals	823	215.340	0.262		
ANOVA excluding “year” factor					
Factor	Degrees of freedom	Sum of squares	Mean square	F-value	p-value
Site	23	44.003	1.913	5.59	0.0000
Depth	2	0.208	0.104	0.30	0.7380
Size	2	2.358	1.179	3.45	0.0322
Site:depth	46	57.873	1.258	3.68	0.0000
Site:size	46	26.418	0.574	1.68	0.0036
Depth:size	4	2.012	0.503	1.47	0.2089
Residuals	935	319.774	0.342		

Table 2. Analysis of variance of $\log(\text{cpue})$ for sablefish from the joint and NMFS long-line surveys (1990–1994). ANOVA includes all main effects and two-way interactions.

Factor	Degrees of freedom	Sum of squares	Mean square	F-value	p-value
Year	4	19.738	4.935	16.72	0.0000
Gear	1	10.837	10.837	36.72	0.0000
Site	23	178.705	7.770	26.33	0.0000
Depth	2	66.757	33.379	113.09	0.0000
Size	2	1971.989	985.994	3340.75	0.0000
Year:gear	4	3.554	0.889	3.01	0.0173
Year:site	92	86.312	0.938	3.18	0.0000
Year:depth	8	6.365	0.796	2.70	0.0060
Year:size	8	28.471	3.559	12.06	0.0000
Gear:site	23	22.132	0.962	3.26	0.0000
Gear:depth	2	0.110	0.055	0.19	0.8301
Gear:size	2	1.176	0.588	1.99	0.1367
Site:depth	46	248.167	5.395	18.28	0.0000
Site:size	46	157.040	3.414	11.57	0.0000
Depth:size	4	105.055	26.264	88.99	0.0000
Residuals	1850	546.012	0.295		

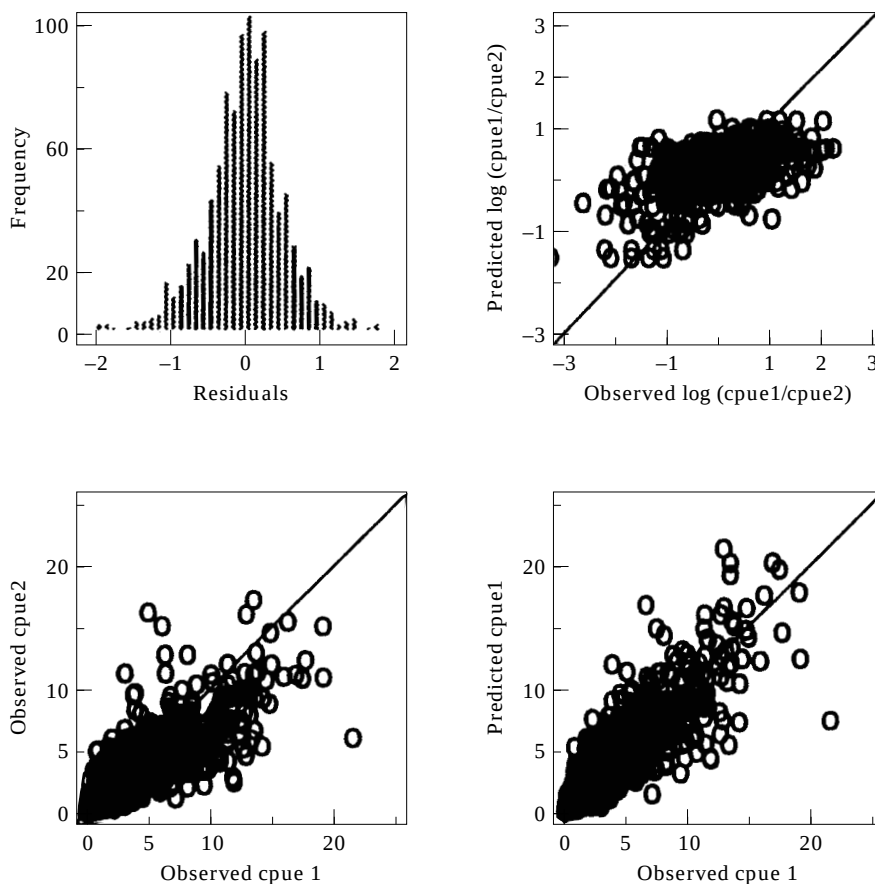


Figure 4. Graphs describing the fit and predictive ability of the log-ratio model. *Top left* shows the residuals of predicted log-ratios from the model fit; *top right* the relationship between observed and predicted log-ratios. *Bottom left* shows the bias when comparing observed NMFS cpue (i.e. cpue1) with observed joint cpue (i.e. cpue2); *bottom right* shows that the bias in joint cpue is eliminated when cpue1 is predicted from cpue2, and compared with the observed cpue1.

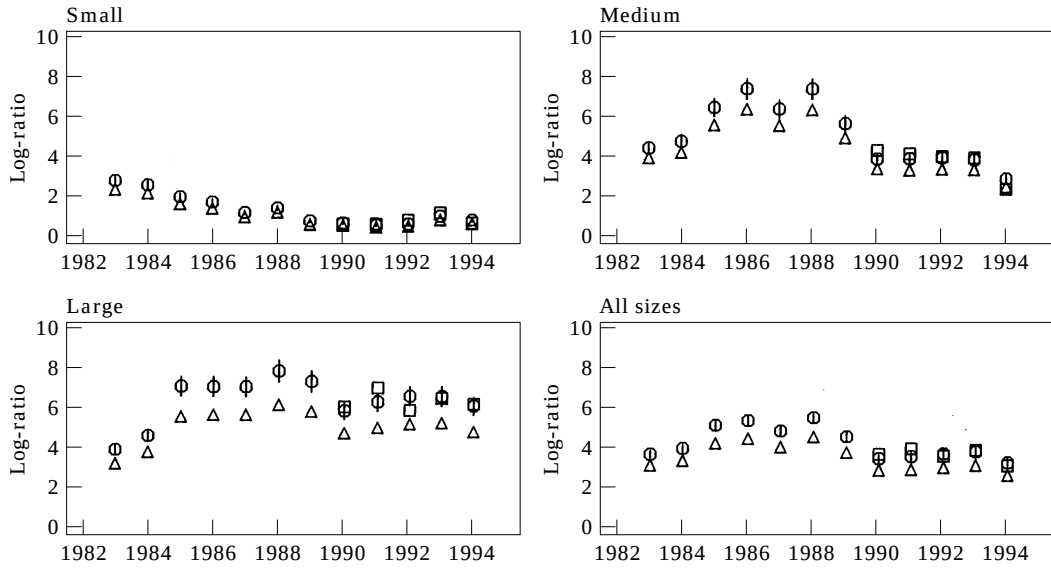


Figure 5. Log-ratio model estimates of relative abundance ($\circ$) made by standardizing joint surveys (1983–1994) to the NMFS surveys (1990–1994). Also shown are the observed average NMFS survey cpue ($\square$, 1990–1994) and the observed average joint survey cpue ($\triangle$, 1983–1994). The bar through the ($\circ$) represents ± 2 S.E. confidence intervals.

depth, size and site:depth and site:size interactions (i.e. we used everything but the depth:size interaction):

$$\log(u_{i1klm}/u_{i2klm}) = \mu + \sigma_k + \delta_l + \rho_m + (\sigma\delta)_{kl} + (\sigma\rho)_{km} + \varepsilon_{iklm}$$

Factors are μ =mean, σ_k =site, δ_l =depth, and ρ_m =size, and site:depth and site:size interactions. The depth main effect must be included in the model even though it is not significant because the site:depth interaction is significant.

The residuals from modeling the observed log-ratios with these factors appear normally distributed, although the plot of observed versus predicted log-ratios appears highly scattered (Fig. 4). The relationship between observed and predicted log-ratios (Fig. 4) illustrates an expected regression towards the mean, because the model is fit to five log-ratios (1990–1994) in each stratum. Comparing the strong bias in observed cpue1 versus observed cpue2 plots with the apparent lack of bias in the predicted cpue1 versus observed cpue1 plots, it appears that modeling was successful in removing the bias in comparing the cpue1 series with the cpue2 series (Fig. 4). Remember that the predicted cpue1 series is estimated from the cpue2 series and the log-ratio model. Relative abundance indices, essentially the joint cpue (i.e. cpue2) standardized to the NMFS survey (i.e. cpue1), with ± 2 S.E. confidence intervals are shown in Figure 5. The component of variability due to sampling the joint survey cpue (Appendix 1) turned out to be negligible when compared to the component of variability due to standardization.

The coincidence of the standardized joint long-line abundance indices with the observed NMFS survey indices for the period of standardization (1990–1994), and for all size categories, suggests that the standardization method based on log-ratios was successful. Only in the large sablefish size category (for years 1991 and 1992) were there instances of significant deviation between standardized joint survey estimates and observed NMFS survey estimates.

We also used the year-effects model with year and gear main effects, and site within gear, depth within gear, and size within gear nested effects to estimate trends in abundance (Fig. 6). As noted earlier, results “standardized” to the NMFS long-line survey are: $\exp(\hat{\mu} + \hat{\alpha}_i + \hat{\beta}_1 + (\hat{\rho}\hat{\beta})_{m(1)})$. We scaled the year-effects relative abundance indices to be on average the same as the log-ratio estimates of relative abundance. Except for the large sablefish category, the log-ratio and year-effects models provided very similar trends in abundance (Fig. 7).

Conclusions

Functional linear regression showed that the relationship between joint survey sablefish long-line cpue and NMFS survey sablefish long-line cpue (averaged within station-depth strata) were similar for the years 1990 through 1994 (Fig. 3). Based on the results of functional regression analysis, we used all of the 1990–1994 comparative survey data to standardize the joint sablefish

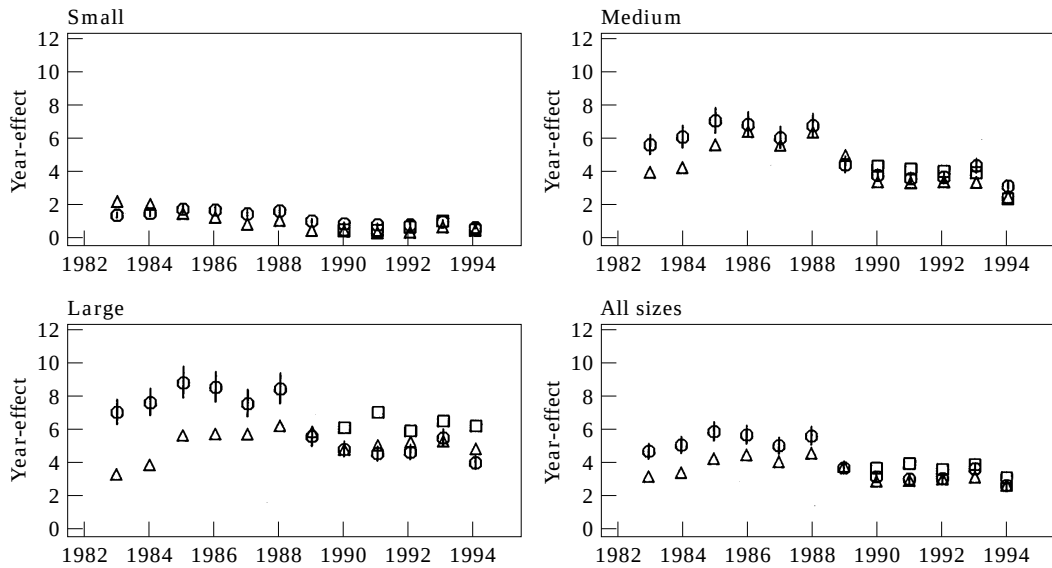


Figure 6. Year-effects model estimates of relative abundance ($\circ$) made by standardizing joint long-line surveys (1983–1994) to the NMFS surveys (1990–1994). Also shown are the observed average NMFS survey cpues ($\square$, 1990–1994) and the observed average joint survey cpues ($\triangle$, 1983–1994). The bar through the ($\circ$) represents ± 2 S.E. confidence intervals.

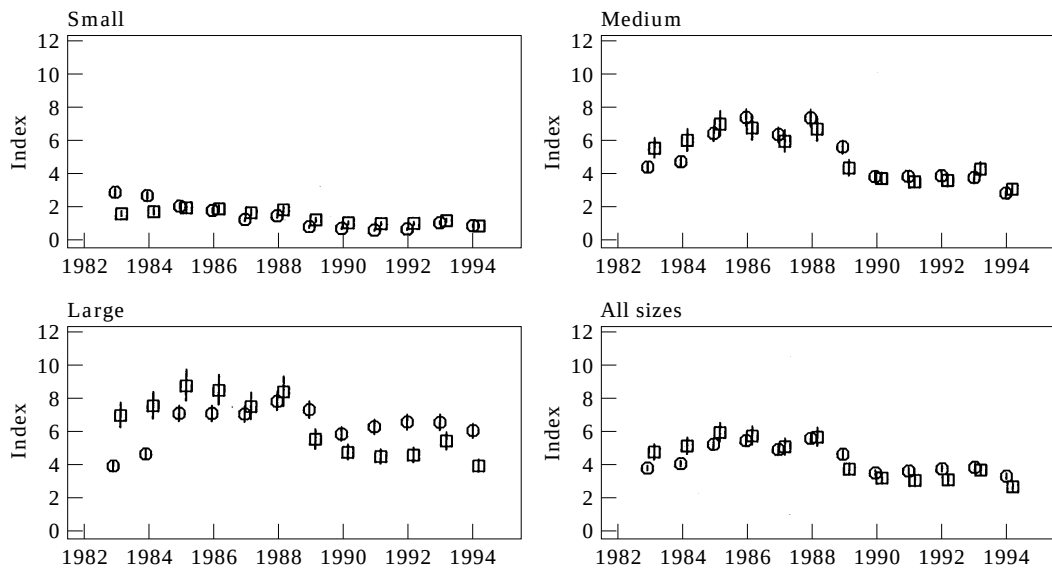


Figure 7. Comparison of log-ratio model ($\circ$) and year-effects model ($\square$) estimates of relative abundance. The bars through the symbols represent ± 2 S.E. confidence intervals.

long-line survey data to that of the NMFS sablefish long-line survey.

ANOVA of log-ratios and log(cpue) data from 1990–1994 indicated that year:gear interaction effects were highly significant (Tables 1 and 2). Obviously, if year:gear interactions are strong enough the data will be impossible to interpret because each gear will display totally different trends in abundance. However, for our data we conclude that this result was partially due to the

large sample sizes available, and that standardization of the joint long-line survey to NMFS survey results could still be meaningful. We feel that most comparative fishing data sets collected over several years would show statistically significant year:gear interactions. Mere statistical significance should not automatically discredit an analysis. What may be more important is that year:gear interactions be smaller than inter-annual changes in abundance as measured by mean square error

(Table 2). Also, from a practical viewpoint, we must consider the merits of performing a standardization against attempting to interpret the observed data without treatment.

Log-ratios appear to simplify the analysis that determines which variables are important to gear standardization. For the sablefish long-line dataset, the ANOVA of log-ratios showed that site, size and the site:depth and site:size interactions were all important factors for predicting log-ratios.

The joint survey sablefish cpues from 1983–1994 were standardized to be comparable to the NMFS surveys using both the log-ratio model and the year-effects model. The coincidence between the standardized joint long-line survey estimates with observed NMFS estimates for 1990–1994, especially for the log-ratio model, suggests that standardization was successful. The coincidence of estimates for these years of standardization also suggest that statistical adjustments for bias were not necessary.

The log-ratio and year-effects models provided similar trends in sablefish abundance. For the selected models, the standardized abundance estimates based on log-ratios appeared to have smaller standard errors than the standard errors based on the year-effects model. This may be due to model selection, and the fact that only paired cpue data are used to estimate parameters for the log-ratio model.

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Appendix 1. Estimating the sampling covariance matrix of the joint survey cpue2, using skate to skate variability of catch per skate, and the results of size category sampling

In order to estimate the variance of the standardized joint long-line survey results, it was necessary to estimate the covariance matrix of joint survey cpue2, stratified by site, depth and size categories (say, $\Sigma(\text{cpue2})$). We had at our disposal individual observations of catch in numbers of sablefish per skate on an individual skate basis, and a size category frequency for each site-depth stratum. Catch per skate for every site, depth and size category were calculated by multiplying the total catch per skate for a particular site-depth category times the proportion of fish in a particular size category. Therefore, for a given site-depth stratum we can drop the subscripts,

$$\begin{aligned} \text{cpue}_{i2klm} &= \left(\frac{\text{total catch}_{i2kl}}{\text{total skates}_{i2kl}} \right) \left(\frac{\# \text{size } m}{\# \text{sampled}} \right) \\ &= \hat{p}_m \end{aligned}$$

and

$\text{var}(\hat{b}\hat{p}_m) = \hat{b}^2 \text{var}(\hat{p}_m) + \hat{p}_m^2 \text{var}(\hat{b}) - \text{var}(\hat{p}_m) \text{var}(\hat{b})$ using Goodman's formula for unbiased variance estimators.

Assuming n_s individual skates are observed with catch per skate X_t , $\text{var}(\hat{b}) = \Sigma(X_t - \hat{b})^2 / [n_s(n_s - 1)]$, and assuming multinomial sampling for the size category we have $\text{var}(\hat{p}_m) = \hat{p}_m(1 - \hat{p}_m)/n_c$ where n_c is the sample size for the size category sampling. The covariance between the

cpues for different size categories in the same site-depth stratum is for example $\text{cov}(\text{cpue}_{i2k11}, \text{cpue}_{i2k12}) = -\hat{b}^2 \hat{p}_1 \hat{p}_2 / n_c$. We can assume that the catch per skate from different site-depth strata are independent.